Selection, Not Capacity

A measured decomposition of retrieval failure in conversational memory, and what survived eleven negative results

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Executive summary

What was tried. Eleven pre-registered attempts to give a language model a working memory of a long conversation: store each exchange, then rebuild a small context every turn. Ten numbered studies plus one retrieval bakeoff, across summarizers, promotion filters, a topic layer, a knowledge graph, a query router, and approximate search.

What survived. None of those mechanisms cleared its own success criterion. What is left is store everything verbatim, keep the recent turns, retrieve by similarity, and pick a set that covers different topics — with no generative model calls anywhere in the memory path.

Four findings.

1. **The context budget was never the limit.** A perfect picker reaches the target using a sixth of the available space. The deployed system spent all of it and delivered a third as much.
2. **For "list everything about X" questions, the most similar records were the least useful.** The four closest matches to the question carried none of its answers. For ordinary lookup questions the same ranking was near-perfect. That split rests on one enumeration question. EC-001 tested the ranking on cleaned LongMemEval-S: only 14.7% of 470 answerable questions put all evidence below the top four, with median evidence rank 2. The internal inversion is not a dominant external pattern. That ranking result does not imply a working retrieval path: 401 questions have evidence in the top four, but only 96 of those retrieve any evidence session after thresholding and packing.
3. **Fix order matters, and the intuitive order is wrong.** The deployed shortlist contained no record at all from one of the four topics, so no selection rule of any kind could cover all four from it. Improving the rule before widening the shortlist cannot work, and measured once, it made things slightly worse than the baseline.
4. **Rank coarse, pack fine.** On 465 externally sourced, turn-labelled questions, strict evidence delivery is 375 with session ranking and session packing, 388 with session ranking and episode packing, and 351 with episode ranking and episode packing. Fine packing helps while fine ranking removes the session context that rescues weakly matching evidence.

Three operational instructions.

- **Do not prune the candidate pool to control cost.** Dropping the least similar records is the one operation measured here to break retrieval — it cost an entire topic even though most of the best records survived the cut.
- **Check whether your memory layer needs model calls at all.** Every component removed here was one that required them.
- **Know your latency horizon before designing for scale.** Storage is free; retrieval time is not.

Cost and risk. Disk is trivial — about 48 MB at ten thousand turns. Retrieval time binds well before that: 190 ms at a thousand stored records, with most of it in one stage whose share is still growing. On this hardware the design is comfortable to a few thousand records and unusable in an interactive loop somewhere before ten thousand.

What this does not establish.

- **The instrument's run-to-run band is 3.0 points on the 13-point rubric, and it is now measured rather than assumed.** Five replicates of the deployed configuration under identical corpus, settings, seed and runtime scored 8.0, 8.0, 8.0, 8.0 and 11.0. **No scored comparison in this arc below about three points is demonstrated** — including the 3.0-point memory-tier contrast this paper cites as its cleanest architectural number. They may be real; a single run per arm cannot tell.
- One internal conversation corpus and one external benchmark, still one model, one machine, one seed, and no error bars. EC-001′s evaluator substitution forbids direct comparison against published LongMemEval systems.

- Every selection count here measures whether a fact was *present in the window*, not whether the model answered correctly. **Those are not the same property, and one live run showed them moving in opposite directions**: the final configuration made six more facts available offline, then scored *lower* than the baseline on questions asking for one specific fact. It is **not promoted**.
- The breadth findings rest on a single enumeration question.

Two images carry the argument between them. Figure 1 is the budget claim: the target is affordable and deployed selection spent everything without reaching it. Figure 3 is the ordering claim. RD-001 recovered the complete 119-episode ordering under the pinned embedding call, so the figure now shows every eligible episode rather than the 16 ranks available when the paper was drafted.

Abstract

Eleven pre-registered attempts to build a memory layer for long conversations produced one architecture worth keeping and a measurable account of why the rest failed. This is a single-program experience report: one internal corpus, one model, one seed, and no error bars. A later external stress test on cleaned LongMemEval-S is reported, but evaluator substitution prevents an official or published-system comparison.

The failures share a cause that can be measured. Write-time selection failed in five studies because it cannot anticipate a later query. Moving selection to query time failed in a registered bakeoff, which reached fewer target facts than write-time formation had. Graph construction, query-type routing, approximate search, query segmentation, and attention-derived term selection each failed their own gates. Underneath all of them sits a candidate ordering that, for this corpus's one enumeration probe, is anti-correlated at the top: the four highest-similarity episodes carry none of its target facts and the last needed item does not appear until rank 87 of 119. The same ordering places every needed item inside rank 2 on all eight lookup probes, so this is one probe behaving unlike eight rather than an established property of query types. On 470 answerable LongMemEval-S questions, the top four ranked sessions contain no evidence for 14.7%, while the median evidence-session rank is 2 and the 95th percentile is 23. The internal inversion therefore does not generalize as a dominant external pattern.

Separating the failure gives three constraints that bind in a forced order, and the order is the paper's operational result. **The candidate pool binds first, structurally**: the art domain has no representative anywhere in the deployed 34-episode pool, so no selection rule of any kind reaches four domains from it — this follows from the pool's contents, not from any measured comparison. With the selector held fixed, widening the pool to 119 episodes moves 5 of 17 items across 2 domains to 12 across 4. **The objective binds second, and only after that**; run on the deployed pool the shipped objective scores 5 of 17 against the 6 of 17 baseline it replaces, one count from one run, illustrating the ordering rather than establishing it. **A similarity floor binds last**: the final missing episode needs a query cosine of 0.225 and has 0.056, which no reweighting closes.

Capacity was never the constraint. An exact optimum on the same store makes 14 of 17 items available in 5,058 of 32,000 characters; deployed selection made 6 available while spending 31,946. These are availability counts measured offline against a planted answer key. Tested live once, the configuration did not promote: its six-item availability advantage produced a one-item gain in correctly attributed answers and a 2.0 loss on targeted probes, failing its own pre-registered no-regression bar.

A separate external granularity analysis isolates two levers on one strict measure. Across 465 turn-labelled LongMemEval-S questions, session-ranked whole sessions deliver answer evidence on 375 items; retaining session ranking but packing episodes delivers 388; ranking and packing episodes delivers 351. The 63 items rescued by coarse ranking have median own-episode cosine rank 46, against rank 10 for the 26 items gained by fine ranking. The observed rule is therefore **rank coarse, pack fine**: use the broader context to score relevance and the narrower unit to spend the context budget. This is a posthoc characterization on an exhausted corpus, not a registered universal law.

A prospective LoCoMo holdout confirms why that scope matters. At a registered 16,000-character operating point, ranking adjacent-turn pairs by their own cosine raises complete exact-evidence delivery from **843/1,098 to 935/1,098** over assigning every pair its session's maximum score: 140 gains, 48 losses, gain/loss ratio 2.92, one-sided exact p=6.19e-12. All six conversations are net positive, and source order reaches only 258. The registered disposition is WORKS, bounded to availability. It does not establish reader correctness or replace LongMemEval's opposite corpus-specific mechanism.

What remains after every one of those removals is an append-only store, a recency window, similarity retrieval, and a coverage objective, with no generative model calls in the memory path — a design that is reproducible and free of generated intermediate text because the removed components were the ones that produced it. That description is accurate about the component; §5.2.4 and §5.2.5 record that the live studies behind these results ran two different rules under the same name, and that neither was a window.

Reading this paper

Terms. The program's vocabulary, defined once.

Term	Meaning here
Episode	One stored user message and its assistant reply, kept verbatim. The unit of storage, retrieval, and cost throughout
Store	Every episode of one conversation. The §5 store holds 119 episodes eligible for its final probe; §6.4's holds 1,000
Probe	A scripted question asked at a known turn, with a locked answer key
Targeted probe	Asks for one specific fact — "who led the survey?" Eight of them
Breadth probe	Asks the model to enumerate everything it knows across topics. **The program has exactly one**, at turn 120, worth 17 items
Domain	One of the conversation's four subject areas: civil engineering, art history, monetary policy, marine biology. The breadth probe's 17 items span all four
Cosine	Similarity between two embedding vectors, 0 to 1. Used two ways below: a **cosine value** (0.056) and a **cosine rank**, an episode's position when the store is sorted by that value (rank 112 of 119)
Availability	Whether an item's text was present in the delivered context. **Not** whether the model answered correctly
Budget	A hard ceiling of 32,000 characters on the delivered context, charged at exact serialized cost including tags and separators
Candidate pool	The episodes a selector is allowed to consider, after any pre-filter
Selector / objective	The rule choosing which candidates to deliver. The deployed one ranks each episode independently; a *set-level* objective scores an episode by what it adds to those already chosen

Work identifiers. Studies are numbered 001–010. Later work carries prefixes: **AR** achievability, **DR** diagnostic repair, **DX** diagnostic, **E** mechanism experiment, **CC** component closeout. The five that matter here:

ID	What it did
AR-001	Computed the cheapest set of episodes satisfying the breadth probe
E005	Swept three set-level selectors over 146 configurations
DR-002	Froze one configuration and varied only the candidate pool
DX-001	Asked why one known-optimum episode is never selected
DX-002	Asked whether a 1,000-turn run's context was still growing

Route. For the argument: §5, then §6. For the methodology lessons: §7.2 and §7.3, which are the most transferable content here. §4 is background.

1. Introduction

A long conversation forces a trade. Keep the transcript and the model slows down and loses the middle. Summarize it and the details are gone. This program spent eleven pre-registered efforts on a third option: store every exchange verbatim and rebuild a small, relevant context each turn.

Most of those efforts failed. This paper is about what the failures had in common, which turned out to be measurable, and about what was left after the failed parts were removed, which turned out to be small.

1.1 The category, declared

This is a case study of one program, not a general-claims paper, and the distinction decides which sentences are permitted. Study 003 retired external baselines in favour of self-comparison. EC-001 later ran cleaned LongMemEval-S, but its substituted evaluator does not authorize an official score or published-system comparison. The internal decomposition still comes from one corpus, one rubric locked since Study 002, one local model at one quantization, one machine, one seed. Every comparison is a single run. There is no significance test that would mean anything, and until Amendment 001 there was no variance estimate either. There is now: **3.0 points**, measured by running the deployed configuration five times under identical conditions (experiments/study_011/noise_band/). Scored differences smaller than that are reported here as *not demonstrated*, in both directions, including the ones this program would rather keep.

Findings below are therefore stated as *on this corpus*, and where a result would be worth testing elsewhere the paper names the experiment rather than implying its outcome. §8 states the limits once, in full, rather than re-stating them at every claim.

1.2 Contributions

1. **A decomposition of where retrieval failure lives** (§5): a candidate pool, a selection objective, and a similarity floor, each separately bounded. **They are not independent — they bind in a forced order**, and applying the second fix without the first makes the shipped configuration worse than the baseline it replaces (§5.6).
2. **The measurement that makes the decomposition possible**: a per-fact known optimum computed on the same store under exact serialized-cost accounting (§5.1). It needs an answer key and exact cost accounting, which is why it is unusual rather than difficult.
3. **A subtraction result** (§6): what eleven efforts removed, and why the properties that make what remains deployable follow from the negative results.
4. **A correction record** (§7), including one instance where a diagnostic written to catch a specific failure class nearly committed that exact failure.

1.3 What this paper does not claim

No claim that any scored difference below about three points in this arc is real. The instrument's run-to-run band is 3.0 on the 13-point rubric (§8.1), so the memory-tier contrast, the live-validation kill and the tier-isolation kill are all reported as measured and none as demonstrated. What survives is the offline decomposition, which is counts and identities rather than scores.

No comparison against HippoRAG, Mem0, Zep, or Letta; none were run. No general claim that similarity retrieval fails — §5.5 measures the opposite on eight of nine probes. No novelty for maximal marginal relevance, facility location, or submodular selection, which are established methods; what is offered is the decomposition and the measurement, not the selector. And no claim that 12 of 17 is good: it sits below the program's registered bar of 14, and below the 15 a known optimum reaches on the same store for a sixth of the cost.

2. Related work

Studies 001–010 were designed and run before this program's first literature scan and are committed with SHAs that show it. We state that once, because it explains why early designs rediscovered known ideas, and press it no further: arriving independently at a worse version of a published method is not a contribution.

Entity-centric indexing. HippoRAG builds a knowledge graph over extracted entities and retrieves by traversal; the authors' own follow-up identifies entity-centricity as a limitation. This program's hardest repeated failure is consistent with that. Six target facts sit in spans where the entity extractor finds zero entities, so an entity-gated index has no path to them — which is why entity extraction was ruled out as a primary index here.

Structure over retrieved units. GraphRAG, SGMem, and CodaRAG impose explicit structure on retrieved material. This program built the corresponding mechanism, an associative graph over observed co-activation, and no configuration cleared its advancement gate (§4).

Deployed systems and benchmarks. Letta, Mem0, and Zep ship in this space; LoCoMo and LongMemEval were the calibration this program lacked. EC-001 has now run the unchanged component on cleaned LongMemEval-S, but its Codex-substituted evaluation is not the benchmark's official score and cannot be compared directly with published systems. LoCoMo remains unrun.

The placement is narrow: every system above consumes a candidate set produced upstream by similarity ranking, and this paper measures that set rather than proposing another structure over it.

3. Method

Pre-registration and binding gates. Each study's design was committed before implementation, and that commit's SHA is the integrity anchor. Offline gates run before inference and bind: Study 008 stopped at its pre-run gates when replay proved no registered fill cap between 1 and 50 could pass the breadth and targeted gates jointly, preventing four invalid 121-turn runs. Amendments never edit a locked registration; they are standalone files recording whether they preceded the result they affect, and the bakeoff carries twelve.

Exact serialized cost. Every character budget charges the complete serialized block — per-episode tags, metadata, and separators included, not just source text. This was not true before DR-001 (§7.2), and correcting it moved published numbers by up to 68%. Every figure here uses the corrected accounting.

Determinism and leakage control. Fixed seed, one slot, speculative decoding disabled, and a required byte-identical seeded-prefix rerun. Mechanism code — retrieval, formation, ranking, gating — may not read the answer key; measurement may. The boundary is enforced by grep, import-graph checks, and a planted test violation that must be caught. Controls run from checked-out prior code in a separate worktree; results from flag-disabled arms were rejected.

4. The arc

Ten numbered studies and one registered bakeoff. The table is the record.

#	Added	Outcome	Terminal diagnosis
001	Recency plus similarity retrieval	PARTIAL, 2 of 3 bars	Similarity fired once in 32 turns. Thirty topics for 32 episodes compressed nothing
002	Consolidation, rule pinning, 120 turns	PARTIAL, 3 of 4	Similarity recovered buried facts; consolidation produced 52 topics
003	Long-term write path, four promotion filters	PARTIAL, 2 of 3	The weighted route was arithmetically unreachable — novelty and association were complementary values from one centroid, capped below threshold — so every promotion used the bypass, making it a novelty-spike detector
004	Long-term read path and arbitration	PARTIAL, 1 of 3	Retrieval ran on all 90 eligible turns with zero displacement; the store lacked the later-domain facts
005	Permissive capture, extractive distillation	PARTIAL	Absolute entity and number counts selected the longest responses. The salience metric was a verbosity detector; 2 of 4 domains formed
006	Length-normalized span selection	PARTIAL, 1 of 3	Formation reached 4 of 4 domains; records shrank about 28×, and count-based retrieval budgets silently broke
007	Character-budgeted retrieval	PARTIAL, 2 of 3	Best of the series. The model used all 10 delivered facts and invented none; 7 required facts were absent from the store
008	Rendering-by-floor factorial	STOPPED AT PRE-RUN GATES	No fill cap from 1 to 50 passed breadth and targeted gates jointly
009	Pure-STM null test	PARTIAL, null decisive	Same seed: 9.0 without the memory tier, 12.0 with it. **The 3.0 gap equals the measured instrument band and is not demonstrated** (§8.1); the baseline was also a locked prefix, not a recency window (§5.2.5)
010	1,000-turn endurance	STOPPED AT G2	Post-stop arms were budget-noncompliant by 67.9% and 68.2%; scores unaudited; one bar not evaluable
—	Retrieval bakeoff	MIXED	Query-time selection did not recover what formation missed

Two results from that table carry the argument forward.

Write-time selection cannot anticipate a query. Studies 003 through 007 are five attempts to decide, at write time, what deserves remembering. Each optimized a proxy satisfiable without the property it certified.

The terminal diagnosis is specific: density, the best write-time salience signal, ranks the six hardest planted facts between 89th and 316th. A later IDF audit did not establish a replacement: its three unregistered variants disagree, and one of the six spans remains ineligible under every variant. These are facts like *photophores* and *ultramarine glaze* — rare technical phrases whose component words are common.

Moving selection to query time did not recover it. The bakeoff registered that repair as its central premise and refuted it. The best registered 32,000-character retrieval block surfaced **8 of 17** target facts, below the 11 of 17 the formation era reached. All 17 were present in the raw store; retrieval did not find them. The bakeoff also advanced none of three further pillars: graph retrieval cleared no advancement gate at any of eight edge types and three depths; query routing had an *oracle* upper bound of 6.09% against a registered 10% build threshold; approximate search degraded recall at synthetic scale.

Scores above are post-audit corrected values (§7.1). They are not a controlled series — runtime and response budgets changed across it — and the only clean architectural comparison the program has is Study 009′s same-seed pair.

Each of these is an unremarkable negative result alone. §5 is what they have in common.

5. The decomposition

5.1 The target was present and affordable

What the counts below measure. 6 of 17, 12 of 17, 14 of 17, 15 of 17 are *availability*: whether an item's text was present in the block delivered to the model. They are not answer correctness. No arm of this selection work was ever scored end-to-end. §5.1.1 records a specific way the two come apart here.

Before asking why selection failed, the program asked what success would have cost. AR-001 computed, on the same store and under exact serialized accounting, the cheapest set of episodes making the breadth items available.

All 17 items are present; 76 of the 119 eligible episodes carry at least one. The exact minimum for 14 of 17 is **5,058 characters across five episodes**, leaving 26,942 characters of the 32,000-character budget unused. A greedy variant reaches 15 of 17 for 5,455. Even 17 of 17 costs 7,592. The most expensive domain, art, needs 3,182 — under a tenth of the budget.

Deployed selection made **6 of 17 available while spending 31,946 characters**.

That figure is packing-conditioned, and §5.2.2 separates the two causes. On the same store, the same candidate identities, and the same selector, offering the similarity hits before the recency window makes **7 of 17 available in 31,863 characters** — one more item for slightly less spend. Part of what this section calls a selection failure is a fill-order failure.

Figure 1. The budget was never tight. Selection spent it on episodes carrying nothing.

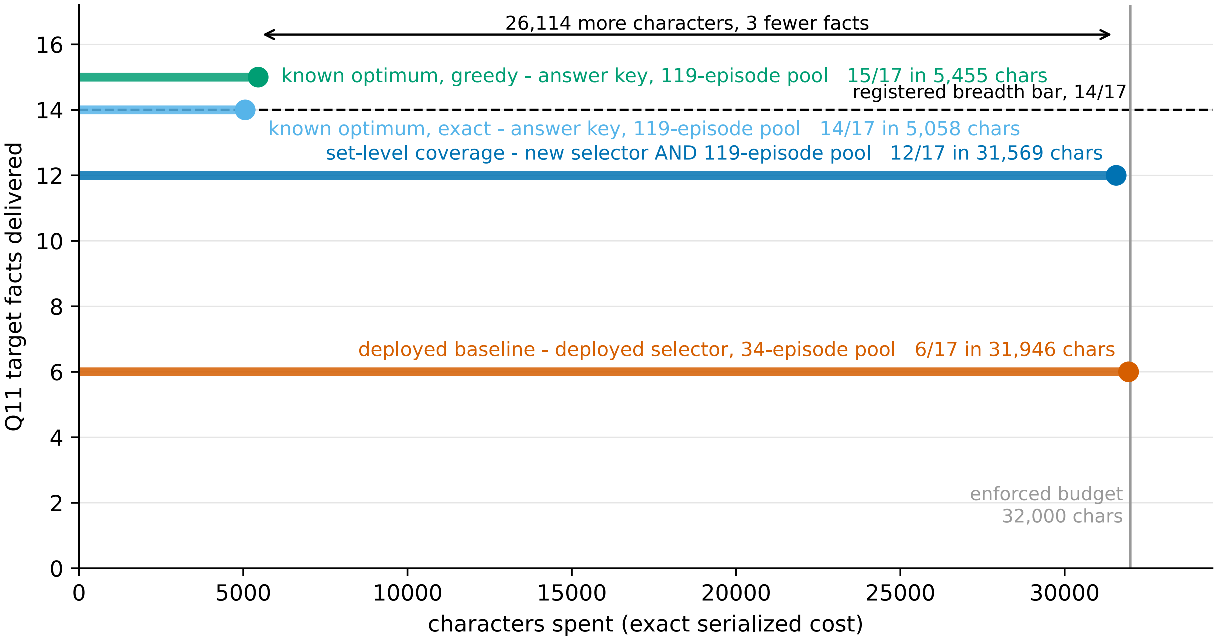

Figure 1. The budget efficiency gap. *The constraint is not capacity: the tallest result is also the narrowest.* Each horizontal stem runs from zero to the characters spent, at a height equal to the facts made available, over the same store at the same enforced 32,000-character budget. The deployed baseline reaches 6 of 17 for 31,946 characters; the shipped set-level configuration 12 of 17 for 31,569; the exact known optimum 14 of 17 for 5,058, leaving 26,942 unused; its greedy variant 15 of 17 for 5,455. **The top two bars change two things against the baseline, not one — the selector and the candidate pool.** §5.3 separates them: on the deployed 34-episode pool the same set-level configuration scores 5 of 17, below the baseline's 6. Read this figure as the budget argument only. Both optima are computed with the answer key and are bounds, not methods. Sources: a0_baseline.json 7645e4746715a965, e005_results.json 07b714389697c6e5, achievability.json 770792d09e07978d.

Two cautions attach to the optimum wherever it appears below. It is computed with the answer key, so it is a bound and not a method. And there are two sets, easily conflated: the exact 14-fact optimum over turns {90, 112, 113, 115, 118} and the greedy 15-fact set over turns {90, 112, 113, **116**, 118}. Everything the program calls "the oracle", including every overlap figure here, is the greedy 15-fact set.

5.1.1 The optimum is mostly the conversation's own earlier answers

Four of its five episodes are prior probe exchanges; only turn 90 is an original plant source. So "the target was reachable in 5,058 characters" partly means the facts had already been restated compactly in earlier answers, and retrieving those restatements is cheap. Achievability holds under the registered eligibility rule — any episode before the probe turn counts — and not under a stricter plant-source-only rule.

There is a sharper consequence, and it bounds how much any count in §5 means. A selector preferring prior answers propagates prior errors, **and this probe's earlier answers were largely wrong.** Availability is scored on an item's presence in the delivered text, so an earlier answer naming the right entity inside an otherwise incorrect response makes that item "available".

Part of the 15-of-17 optimum, and of any configuration scoring well by recovering those four episodes, therefore consists of delivering the conversation's own earlier mistakes with the correct nouns in them. The decomposition survives this — it concerns which episodes get selected, not whether they are true — but "makes 15 of 17 available" and "would help the model answer correctly" are further apart than an availability count suggests.

That distance has since been measured once, and it is large. LV-001 ran the shipping configuration and the deployed baseline live over this corpus at one seed. The six-item offline availability advantage became a **one-item** difference in correctly attributed answers, while targeted accuracy moved the *other* way: the shipping configuration scored 1.5 of 8 against the baseline's 3.5. Asked for the two formatting rules planted in turns 1 and 2, it reported that it could not see the start of the conversation at all — the coverage objective had spent its budget on domain spread. Offline, the same configuration preserved 16 of 16 targeted items. Availability was preserved; answers were not. §8.6 carries the consequence.

5.2 The candidate pool binds first

Selection runs over whatever the pre-filter admits. DR-002 froze the shipped configuration and varied only pool membership — same store, same renderer, same embedding.

Pool	Candidates	Facts	Domains	Known-optimum overlap
deployed pre-filter	34	5/17	2/4	1/5
cosine top-100	100	9/17	3/4	0/5
full eligible store	119	12/17	4/4	4/5

Widening the pool moves the same configuration from 5 of 17 across 2 domains to 12 across 4. Figure 2.

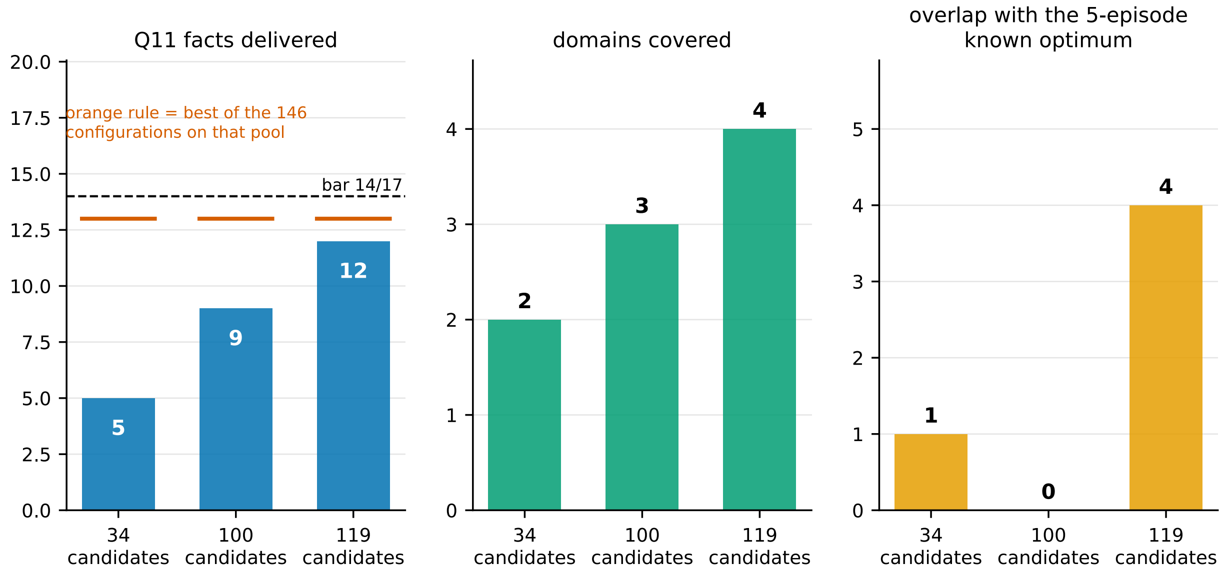

Figure 2. Pool ablation. *Widening the candidate pool from 34 to 119 episodes, with the selector frozen, moves the same configuration from 5 of 17 across 2 domains to 12 of 17 across 4.* Facts, domains, and known-optimum overlap at three pool sizes, everything except pool membership held fixed. Dropping only the 19 lowest-cosine episodes to form the 100-pool costs three facts, the whole art domain, and all optimum overlap, though four of the five optimum episodes survive the cut — the selector clusters over the pool, so tail removal reshuffles the objective rather than removing options. The orange rule marks the best of 146 configurations on each pool: 13 of 17 on all three, which is why the pool's binding effect must be read in domain coverage. Sources: configuration_sweep.csv 1ad625d10fb988f9, pool_secondaries.csv 5987d05846c64f97.

Two things keep this honest. It is a **frozen-configuration** readout, not a sweep: the best of 146 configurations reaches 13 of 17 on all three pools, so the pool's binding effect is properly read in domain coverage rather than in that maximum. And on the deployed 34-episode pool **no configuration covers four domains at all**, because the art domain has no representative anywhere in the top 34. That is a statement about what the pre-filter makes possible, not about how well any selector searches.

The middle row deserves a second look. Dropping only the 19 lowest-cosine episodes to form the 100-pool costs three facts, the whole art domain, and *all* overlap with the known optimum — though four of the five optimum episodes survive the cut. The selector clusters over the pool, so removing the tail reshuffles the objective rather than merely removing options. Pool size does not predict what removal costs, which is why §6.4 rejects pool trimming as a cost control.

5.2.1 Why the pool binds: the ordering is inverted at the top

DR-002 registered its rule before computing any rank: *any fact-bearing selection at cosine rank 80 or worse means cosine ordering is the wrong prior for breadth.*

It fired. The worst fact-bearing selection sits at rank 86 and carries two of the four art items. Around it:

- **The four highest-cosine episodes in the store carry zero target facts.**
- The first fact-bearing episode is at rank 5; the last still-needed item does not appear until rank 87 of 119.
- Both art contributors sit at ranks 50 and 86 — which is exactly why the deployed 34-episode pool cannot reach four domains.
- The five known-optimum episodes sit at ranks 14, 20, 22, 86, and 112. The shipped configuration recovers four, including the one at rank 86, and misses only the deepest.

Figure 3. The failure is not that the ordering is noisy. At the top it is anti-correlated: the most query-similar episodes are the least informative for this probe, and one of the two domains deciding the gate is unreachable before rank 50. That a set-level objective reaches rank 86 at all bounds the claim — the objective partially compensates for a bad ordering, and does not compensate fully.

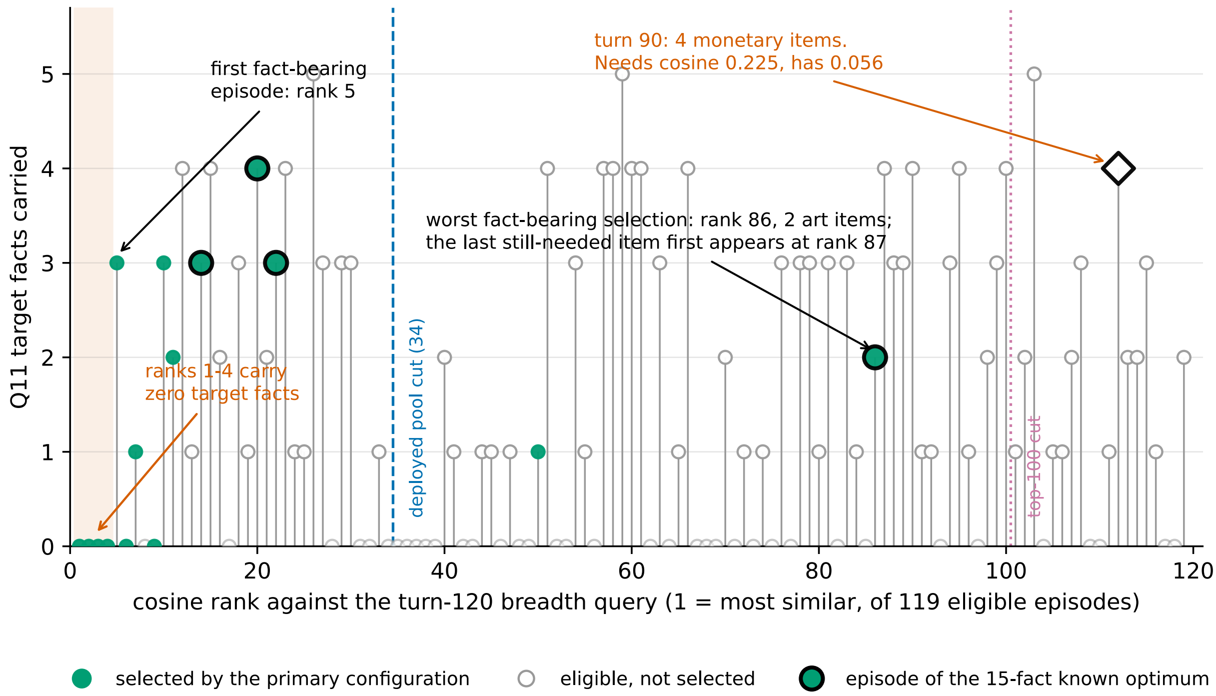

Figure 3. Cosine rank against fact content. *On this corpus, the enumeration probe's target facts sit outside the top of the cosine ranking, and the deployed pool cut removes an entire domain.* Horizontal axis: cosine rank against the turn-120 breadth query over the 119 eligible episodes. Vertical axis: target facts carried. The four highest-ranked episodes carry zero; the first fact-bearing episode is at rank 5 and the last still-needed item does not appear until rank 87. The five episodes of the 15-fact known optimum are marked at ranks 14, 20, 22, 86 and 112. Both art contributors lie at ranks 50 and 86, so the deployed 34-episode pool contains no art episode and cannot reach four domains at any setting; the 100-episode pool excludes the rank-112 episode carrying four monetary items. The first version plotted only the 16 ranks then available. RD-001 now plots all 119 after replaying the pinned E005 embedding call (§8.8). Sources: full_rank_inventory.csv 7d8874f54d8e9729, q11_selection.jsonl 71d7d1a6f4d46d23, cost_comparison.csv 1ca40da99315c719, and generality_batched.json 7e1fa13ef71a8077 rank 20 supersedes the published 21 per ERRATA.md, 2026-08-01.

The external test narrows this claim. EC-001 ranked annotated evidence sessions for all 470 answerable cleaned LongMemEval-S questions under the unchanged embedder. The top four contain no evidence for 69 questions (14.7%), not as a general rule; across 890 evidence sessions, median rank is 2, p95 is 23, and the maximum is 49. Multi-session questions are not the exceptional case: 13 of 121 (10.7%) have no evidence in the top four. The internal rank-87 pattern remains an exact fact about this corpus and probe, but it is not a dominant property of naturalistic conversational retrieval under this test.

The same external result exposes a different failure downstream of ranking. Session rank is diagnostic; the component thresholds and packs exchange episodes. A post-run audit of the committed mechanism log finds evidence in the top four on 401 of 470 questions, but any evidence-session recall on only 96 of those 401. At least one exchange clears K = 0.48 on 232 of 500 questions, yet a non-recency K exchange survives packing on only 20. Every block is truncated; the median composition is 16 recency exchanges, zero non-recency K exchanges, and one coverage exchange at 31,920 characters. Of 109 session hits, 91 come from recency and only 18 from non-recency paths. The threshold matters most for single-session preference, whose median best-evidence cosine is 0.399, but the dominant observed gate is N-first budget exhaustion. EC-001 therefore narrows the inversion claim without validating the retrieval path that follows the ranking.

EC-002 tests that gate directly. In an offline same-store replay it holds the vectors, K = 0.48, selector, and 32,000-character budget fixed and changes only packing order from recency-first to K-first. Any evidence-session recall rises from 109/470 to 261/470: 152 paired gains and zero losses. Exact-turn-any availability rises from 79/470 to 196/470, with 119 gains and two losses. Delivered K episodes rise from 26 to 476 while every block remains truncated. Packing priority is therefore a causal gate for recovered EC-001 items, not merely a post-hoc correlate. The replay is availability-only and does not authorize production promotion or establish reader-level gains.

5.2.2 The same gate is closed on the internal corpus

Every study in this program's record ran recency-first, so whether the internal corpus was starved the same way was unmeasured until IC-001 replayed the corrected 121-turn run under both orders. It holds the store, the committed candidate identities, K = 0.48, N = 32, the selector, the budget, and the renderer fixed, re-derives no vector, and changes only fill order. Its B0 arm reproduces the deployed 6-of-17 result exactly — same characters, same eight episodes, same payload digest — before the K-first arm was opened.

The expected effect was smaller here. One continuous conversation gives recent turns and relevant turns far more overlap than EC-001′s discontinuous histories, so recency-first should waste less. The breadth effect is indeed small: 6 of 17 to **7 of 17**, one gain and no losses. The delivery effect is not small at all.

	Recency-first	K-first
Probes where the K path delivered nothing	**8 of 8**	3 of 8
K episodes delivered across the eight probes	**0**	9
Q11 episodes delivered	8	12
Q11 characters spent	31,946	31,863

Under the order this program shipped for eleven studies, the similarity path delivered **zero episodes and zero characters at every probe**. Recency consumed the whole budget every time, exactly as EC-001′s external audit found. Admitting two small similarity hits first then left room for four *more* recency episodes than the deployed order fitted, because the deployed walk had already spent the budget on one large episode — so the K-first window is larger and cheaper at once. It is a priority effect under a binding budget, not extra capacity, and on this corpus it is not a trade: no targeted probe fell, two rose from 0 of 2 to 2 of 2, and the window gained the turn-1 and turn-2 episodes whose absence LV-001 reported live.

The consequence for this section is narrow and real. The 6-of-17 baseline is a joint readout of a selector and a fill order, and §5.3′s within-pool selector comparisons inherit that: they compare objectives that were all run behind the same starved packing order. IC-001 is availability-only on one probe, one store, one run, with no variance, and authorizes no re-run of the arc.

5.2.3 The gate was real, and opening it did not help

Study 011 ran the arc live to find out. Four 121-turn runs at one seed — recency alone, similarity alone, both with similarity first, and both in the deployed order — scored blind by three raters who never saw which arm produced which answer.

The suppression is confirmed at full strength. **The deployed arm scored identically to the recency-only arm on all thirteen questions**, with the same availability on both measures, and produced byte-identical windows at three consecutive late probes. A system carrying a similarity tier was indistinguishable from one with no similarity tier at all.

Then the correction was applied, and the answers got worse.

	Recency only	Deployed	K-first
K-path episodes delivered	0	1	**13**
Q11 items available	9 of 17	9 of 17	**10 of 17**
Targeted items available	7 of 21	7 of 21	**10 of 21**
Scored rubric, out of 13	8.0	8.0	**7.0**

The K-first arm had the best availability of any arm and the worst score. Its registered kill — *the corrected arm must not score below the deployed one* — fired, and the correction was not adopted.

The shape matters more than the total. The point is built from three gains and three losses, not a uniform decline: the gains are the early and middle plants, where the similarity path had material to contribute, and every loss is late. Both losses at the marine-biology probe fall on a turn that holds **no similarity candidate at all**, so nothing the similarity tier did could have helped there; what changed was the recency context it displaced elsewhere. The mechanism is consistent with displacement and is not established — a single run per arm cannot separate it from ordinary variation in a live conversation.

One caveat governs every scored number above. Re-running a single arm under identical settings produced a byte-identical prompt and a **different answer** at the same seed, with one slot and speculative decoding off. The mechanism reproduces exactly where it can be tested — but that is one turn, since a differing answer changes the store and every prompt after it. A one-point gap on a thirteen-point rubric, from one run per arm, therefore sits inside a noise band this program has never measured. The registered bar fired on the committed numbers, which is what a pre-registered kill is for; the defensible reading is that the correction did not demonstrate an improvement, not that it is worse. The delivery and packing numbers are offline and reproduce exactly.

5.2.4 The tier the arc calls recency is not one

Mechanism analysis after Study 011 was unsealed found that the tier every study above calls a recency window does not select by recency. Its ordering key sorts the **whole store** by delivery history — never-delivered material first, then the episode delivered longest ago — with the source turn entering only as a third-level tiebreak, oldest first. It is a least-recently-delivered coverage rotation. The only place recency appears is the name of the block it renders into.

Replaying the deployed key against store state reconstructed from the delivery log reproduces the live ranking on **120 of 120 testable turns** in every arm that has the tier. On that licence: the delivered set overlaps a true window of the same size by **0.29**, **36%** of deliveries are older than the cap of 32 turns and so lie outside anything a window could reach, and the rotation touches **every one of the 120 reachable episodes**, which a window never does.

Two measurements explain how this survived eleven studies. The candidate list equals a recency window on exactly the 32 turns where the store still fits inside the cap, and never after — so at ablation length the tier genuinely *is* a window. And it delivers the immediately preceding turn at 9 of 9 probes, because that episode is the one thing never delivered before. The block opens on the previous turn and then rotates through the archive.

Three distinct rules carry the name in this repository, and they are not variants of one mechanism:

Path	Cap	Orders by	Where it ran
RetrievalEngine, StmRetrievalEngine	10	most recently delivered first	Every live run through Study 010
logical_n_key	32	least recently delivered first	Corrected Tier 6 and Study 011
episodic library	32	the last N in conversation order	The extracted component; EC-002, CC-003, CC-005

The first row's span was corrected on 2026-08-08. An earlier version of this table placed Study 010 in the second row. Study 010 ran src/study/runner.py, which constructs RetrievalEngine its arms replay exactly under the first row's rule and do not replay under the second. The two carried engines declare the N tier separately but behave identically, which a probe on shared inputs confirms rather than assumes.

The only genuine recency window is in the component no scored live study ran. The harness imports the library's packer and renderer, not its context composition. §6.2′s description of what remains is accurate about that component; what does not hold is the implied continuity between it and the measured arc.

What this does and does not change. Every contrast in which both arms carry the tier is untouched, which includes §5.2.3′s C-against-D packing result and every delivery and packing number in this paper. What changes is what the *marginal* contrasts mean: the similarity tier was being asked to add to a baseline that already reaches the entire store, not to a window over the last few turns — the most plausible reading of why 79% of the K-first arm's similarity candidates were material the other tier had already nominated. §5.2.5 does the same analysis for the first rule in the table above. **Nothing here establishes what a correctly-implemented recency window would score, in either direction.**

For this paper the correction is narrow. §5.2.2′s delivery finding stands unchanged and is now confirmed live. What does not survive is the inference a reader would naturally draw from it: that the starved fill order was holding back a working retrieval path. On this corpus, at this seed, giving that path the window space it had been denied moved availability up and answers did not follow. The pool, the objective and the floor still bind in the order §5.6 gives them. Packing order gates delivery, and delivery was not the thing limiting answers.

5.2.5 The rule that ran before it was a locked prefix

The rule in the first row of §5.2.4′s table has a third component the key does not show. `retrieve()` refreshes every episode it delivered, in one call, with one timestamp. Since the rule ranks the freshest delivery highest, what is in the block is the freshest thing in the store, so it is selected again, so it is refreshed again. The batch write leaves its whole set tied, and the tie breaks toward the order the store query returns, which is oldest first.

The block therefore settles on the oldest episodes in the conversation and cannot leave them. Replayed against the committed logs it reproduces the ranking exactly — 120 of 120 turns for Study 009′s Arm S, 120 of 120 for the Study 007 arm carried in as its comparison, 34 of 34 for the ablation. From turn 11 both arms delivered **source turns 1 through 9** plus whichever episode had not been delivered before, which is always the immediately preceding turn, and they held that for 111 consecutive turns.

	Value
Mean overlap with a true window of the same size	0.205
Deliveries older than the cap of ten turns	82.6%
Mean age of a delivered episode	53 turns
Episodes delivered exactly once	111 of 120

At turn 120 the block held source turns 1–9 and 119; a last-ten window would have held 110–119. The first episode of the conversation was delivered on all 120 turns; episodes 10 through 118 were delivered once each, on the turn after they formed, and never again. Across the committed record the scan replays 17 run directories exactly, of which 12 lock; **every scored live run from Study 004 through Study 010 is among them**, including Study 010′s arms, which held source turns 1–9 across 999 logged turns.

The two carried arms hold the identical block turn for turn, so the 3.0-point architectural contrast is not confounded and is not re-read here. What the measurement corrects is the baseline's description: that contrast is not a win over a recency baseline but over one slot of genuine recency in ten, above a frozen prefix of the conversation's opening. The same correction reaches the earlier 11.0–7.0 STM-versus-LTM comparison, whose arms carried the same prefix.

The two failures are opposite and nothing transfers between them. The deployed rotation reaches every episode and duplicates the similarity tier; the carried prefix reaches almost nothing and repeats itself. Both were called a recency window, and in both the name was the only thing asserting it.

5.3 The objective binds second, and only after the pool

The deployed rule ranks each episode against the query independently and takes the best until the budget fills. That rule cannot represent redundancy: an episode's value does not depend on what is already selected.

E005 replaced it with three set-level objectives — maximal marginal relevance, facility location, and relevance plus cluster diversity — swept over 146 configurations per pool at the enforced budget, with zero inference calls and a byte-identical rerun. The best gate-passing configuration makes **12 of 17 items available across 4 of 4 domains at 31,569 characters**, preserving all 16 targeted items and recovering 4 of the 5 known-optimum episodes.

That headline moves two variables at once. The deployed baseline is the deployed selector on the deployed 34-episode pre-filter; the 12 of 17 is a set-level selector on the full 119-episode store. Selector and pool both changed, so it is not a selector result and this paper does not use it as one. The per-pool minima across the same 146 configurations make the confound concrete:

Pool	Worst of 146	Best of 146	Frozen shipped config	Deployed baseline
deployed pre-filter, 34	**4/17**	13/17	**5/17**	6/17
cosine top-100	5/17	13/17	9/17	not applicable
full eligible store, 119	7/17	13/17	12/17	not applicable

The baseline column has one entry because the baseline has no pool variable: it is the deployed pipeline's pre-filter and selector together, never run against a wider pool. The blanks are not zero measurements.

Read the first row. On the pool the system actually used, set-level configurations fall as low as 4 of 17, and **the configuration this program ships scores 5 of 17 there — below the 6 of 17 baseline it replaces.** The selector is not a free improvement. It requires the wider pool, and without it is a regression.

The honest within-pool statement is the third column against the fourth: holding the deployed pool fixed, the best of 146 set-level configurations makes 13 of 17 available against the baseline's 6. That is a selector effect, on one pool.

Three further results matter more than the headline.

The highest-scoring selector was the worst one. Facility location reached 13 of 17, the highest raw count, and passed no gate at any setting, because it delivered the monetary domain 0 of 4 every time. It improved the total by abandoning a domain. Only the per-domain check caught it; a reader looking at fact counts would have shipped it. Figure 4.

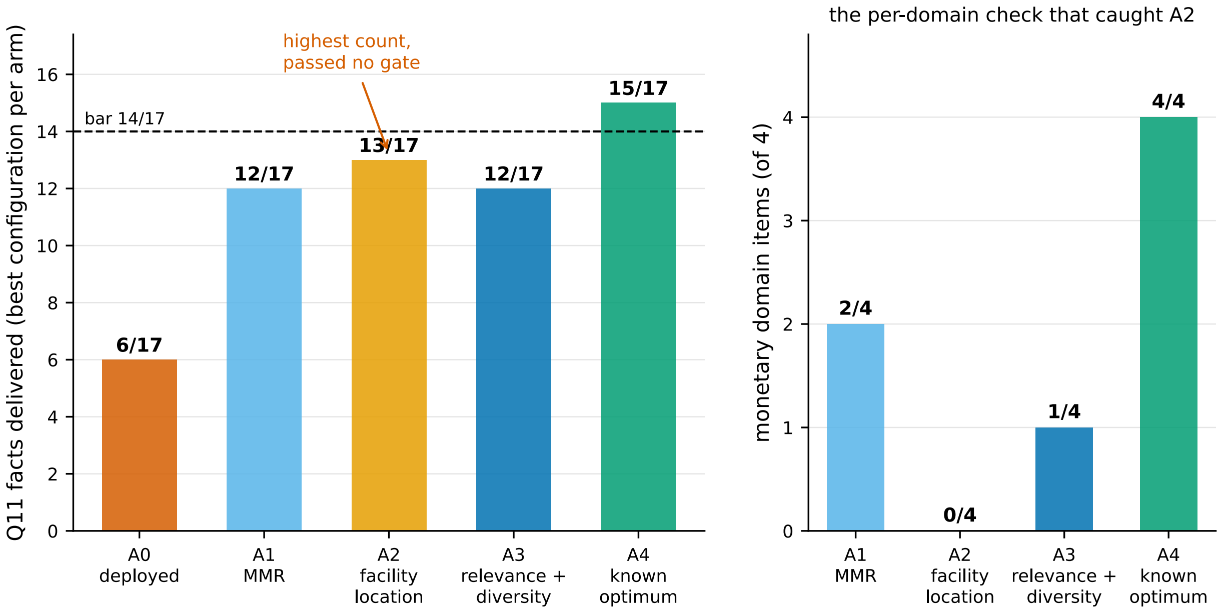

Figure 4. Selector comparison. *The selector with the highest raw fact count delivered nothing from one domain and passed no gate.* Best configuration per arm on the 119-candidate pool at the enforced budget, with the monetary domain broken out. Facility location leads on count at 13 of 17 and delivers monetary 0 of 4 at every setting; the shipped relevance-plus-diversity configuration reaches 12 of 17 across all four domains. On this pool all 146 configurations beat the deployed 6 of 17; on the deployed 34-episode pool they fall as low as 4 of 17 and the shipped configuration scores 5 of 17 (§5.3). 137 configurations preserve 16 of 16 targeted items, so targeted recall does not separate the arms — itself the finding. Sources: configuration_sweep.csv 1ad625d10fb988f9, a0_baseline.json 7645e4746715a965, e005_results.json 07b714389697c6e5, achievability.json 770792d09e07978d.

A parameter registered as inert was not. Cost scaling was predicted to make no difference on a budget with slack. The budget is slack for the optimum and not for a selector registered to fill it, and the prediction failed.

The search is not the limit. The bound here is data-dependent, not a worst-case constant: at the final greedy set, each unselected candidate's marginal gain per unit cost is computed, the remaining budget filled fractionally in that order, and the result added to the achieved objective value to bound the optimum. On the primary 119-episode pool, greedy sits between **0.9548 and 0.9996** of it across the 135 configurations where the bound is computable; the shipped configuration is at 0.9927. A better search over the same objective has almost nothing left to find.

Two scope notes, because this number is easy to quote wrongly. Widening the population to all three pools — 405 computable configurations — moves the minimum to 0.9536; the program's ledger reports 0.955, which is the primary-pool minimum to three places and is the figure this paper uses. Neither value supersedes the other and no published number changes; they count different configurations. And the bound exists only for the two arms with a set function. All 33 non-computable rows are the maximal-marginal-relevance arm, so this reading covers facility location and relevance-plus-diversity and says nothing about MMR's search quality.

5.4 The similarity floor binds last

After the pool is widened and the objective replaced, one known-optimum episode remains unselected: turn 90, carrying four monetary items — the reason monetary is the shipped configuration's weakest domain at 1 of 4.

DX-001 ran a replay gate first, reproducing 146 of 146 committed payload hashes byte-for-byte before reporting anything. The gate earned its place; §7.4 records what it caught.

No configuration in the registered space selects turn 90. That count is weaker evidence than it looks and the paper does not lean on it: the 146 configurations are a grid over three parameters, so sweeping the diversity weight eleven times is closer to one chance repeated than to eleven chances. The arithmetic is the argument.

The registered prediction was cluster collision — the episode shares a cluster with a selected one, so the diversity term goes unpaid. That prediction is **wrong**, refuted twice over: the episode's cluster is never entered by any selection, so the diversity term was payable in full at all 15 steps, and a counterfactual paying it in full wins at no step.

What remains is arithmetic. To win at its best step the episode needed a query relevance of **0.225**. It has **0.056**. Only 20 of the 119 episodes clear that bar, so it would have to be a different episode by cosine, not a better-weighted one. Across 132 walks of the parameter space its best rank anywhere is 4, never 1.

This is a floor. No reweighting of an objective built on that similarity reaches it, which is why the registered no-change branch fired and 12 of 17 ships with the miss characterized rather than tuned away.

5.5 What the inversion does not explain

§5.2.1 invites a larger claim — that every mechanism in this program, and much published architecture in this space, ran downstream of a broken candidate ordering. DR-002 tested that reading and **measured it false**.

On the eight targeted probes, cosine ordering places every needed item **inside rank 2**, and the top four candidates carry a target item on every one. That is near-optimal, and it is why targeted recall runs at 60 of 60 and why 137 of the 146 configurations preserve 16 of 16 targeted items without effort.

The inversion also fails to explain most of §4. Not the formation-side failures, which ran at write time upstream of any retrieval filter. Not Study 003′s arithmetically unreachable promotion route. Not Study 007, where the model used all 10 delivered facts and seven required facts were simply absent from the store. It is contradicted by the bakeoff's routing oracle, which assumed perfect selection and still ceilinged at 6.09%, and by widened raw short-term memory, which delivered all six formation-blind facts — the ones density missed and the later three-variant IDF audit did not consistently rescue — with no selection filter at all, the model using five correctly.

One limit decides how much §5.2.1 can carry: the internal comparison is **eight probes against one**. The program has exactly one enumeration question, so the top-four-zero and rank-87 readings remain a single instance. EC-001 adds 121 multi-session reasoning questions as the closest external breadth analogue, but they are not literal enumeration probes. They score weakly — 24.0% any-turn availability over all 121 and 0 of 20 correct in Tier 2 — while their top-four failure rate is only 10.7%. Breadth weakness therefore generalizes more readily than the inversion proposed to explain it.

So the claim is what was measured: **on this corpus, one enumeration probe behaves completely unlike the eight lookup probes, and the mechanisms this program built for breadth all ran downstream of the ordering that probe exposes.** It unifies the internal breadth failures. LongMemEval shows that multi-session weakness can occur without a dominant top-four inversion, so the ordering does not describe the external category.

5.5.1 What the corpus-artifact test now requires

The program's own description of its hardest facts is that they are rare technical phrases whose component words are common. That is a *lexical* property, and it predicts the observed effect directly: an embedding will place such a span far from a query phrased in ordinary words. If that is the mechanism, §5.2.1 is a finding about this corpus's planted vocabulary and generalizes only to corpora whose target facts are similarly distinctive.

RD-001 pre-registered that discriminating measurement: correlate each fact-bearing episode's cosine rank against the lexical rarity of its key phrases. It recovered all 119 ranks under the pinned E005 embedding call, but then stopped before computing a coefficient. The earlier rarity artifact does not score the registered population: it has three variants for only 6 of the 76 fact-bearing episodes, with no primary variant and no phrase-to-episode aggregation. The other 70 have no unchanged committed rarity score.

The three variants also do not support the paper's earlier shorthand that "IDF ranks them worse." Mean content-word IDF ranks all five eligible hard-plant spans worse than density; maximum IDF improves two of five, and summed IDF per word improves one. The sixth span is unranked because the audit retained an entity-or-number eligibility filter. No variant was designated primary. The categorical claim is corrected in ERRATA.md.

This is a measurement failure, not a null. Choosing a variant, extending it to 70 episodes, or defining an aggregation after the decision rule would decide the result through unregistered choices. No Spearman coefficient or confidence interval was computed, no registered branch fired, and the vocabulary alternative's cause remains unresolved. EC-001 supplies the other registered route: on a corpus this program did not construct, only 14.7% of answerable questions put all evidence below the top four. That narrows the internal inversion to a non-dominant pattern without identifying whether vocabulary, query form, history structure, or ranking granularity caused the difference. The full rank recovery strengthens the descriptive ordering in Figure 3; it does not tell us why that ordering occurs.

Completing the test needs no embedder replay, but it is not free recovery. It requires a prospective RD-002 that fixes one rarity formula and one phrase-to-episode aggregation, then computes 70 new corpus-statistic rows. That design would be registered after the cosine ranks were known, a weaker epistemic position than the original paper implied and one the result would need to state.

5.5.2 Ranking and packing units have opposite effects

NF-002 and NF-003 leave three arms on the same 465 turn-labelled LongMemEval-S items, under the same 32,000-character budget, skip-on-overflow policy, and strict answer-episode outcome:

Ranking unit	Packing unit	Strict delivery
Session	Whole session	375/465
Session	**Episode**	**388/465**
Episode	Episode	351/465

The one-factor contrasts have opposite signs. Holding session ranking fixed and packing episodes gains 17 and loses 4, net +13. Holding episode packing fixed and replacing the inherited session score with each episode's own cosine gains 26 and loses 63, net −37. The deployed middle corner is the observed optimum: **rank coarse, pack fine**.

The discordant ranks identify why. The 63 answer episodes carried by session ranking and dropped by episode ranking have median own-cosine rank 46 and p90 135. The 26 moving the other way have median rank 10 and p90 21. Session pooling supplies contextual evidence for an answer episode whose own text is a weak query match; episode packing then avoids paying for the whole session that supplied that cue. This turns the earlier H2 residual into visible mechanism: coarse context can rescue evidence below the practical fine-rank frontier.

The result is a posthoc synthesis on a corpus whose items are exhausted, so it does not establish a universal direction. In particular, it does not show that coarse ranking wins when the budget admits most candidates. Development-only budget sweeps on LoCoMo and LongMemEval rejected binding ratio as a portable moderator before the external holdout was opened.

NF-004 then registered the corpus-specific opposite direction on six sealed LoCoMo conversations. At 16,000 characters, pair ranking raises complete exact evidence from **843/1,098 to 935/1,098**, with 140 gains, 48 losses, ratio 2.92, and one-sided exact p=6.19e-12. It also stays positive at the secondary 32k point, 961 to 1,024; source order reaches only 258 at 16k. All six conversations are net positive. This passes the prospective WORKS bar and demonstrates that ranking granularity is corpus-scoped under the tested instrument. The endpoint is availability, not reader correctness, and authorizes no live or adoption claim.

5.6 The three constraints, and why the order is forced

Constraint	Binds on	Bound, on this corpus
Candidate pool	domain coverage, and part of the fact gap	The art domain has no representative in the deployed 34-episode pool, so no rule of any kind reaches four domains there. With the

Constraint	Binds on	Bound, on this corpus
		selector frozen, 34 → 119 candidates moves 5/17 across 2 domains to 12/17 across 4
Selection objective	the remaining recoverable facts	Holding the deployed pool fixed, the best set-level configuration reaches 13/17 against the baseline's 6/17. Greedy runs at 0.9548–0.9996 of a data-dependent bound on the primary pool, so the objective and not the search is the limit
Similarity floor	the irreducible residual	0.056 against the 0.225 required, a shortfall of 0.169; only 20 of 119 episodes clear the bar; unreachable by any reweighting of this objective

The order is forced, and the argument for it is structural rather than numerical. **The art domain has no representative anywhere in the deployed 34-episode pool.** Its two contributors sit at cosine ranks 50 and 86 (§5.2.1), so no selection rule of any kind — set-level, per-item, or hand-written — can deliver an art item from that pool. Four-domain coverage is not merely hard there; it is unavailable. Objective work cannot recover a domain the pre-filter never admitted.

The pool has to be widened first. That follows from what the pool contains, not from any measured comparison, and nothing about a single run or a single seed bears on it.

A measured result points the same way and is worth stating for its vividness, with its fragility attached: run on the deployed pool, the configuration this program ships scores 5 of 17 against the baseline's 6 of 17 — the objective fix, applied alone, is a small regression. That is **one count, from one unreplicated run**, and a reader is entitled to weigh it lightly. Note also that the best of 146 configurations reaches 13 of 17 on that same pool, so the shipped cell is not the only cell one could point at.

Neither observation is load-bearing. **The structural claim carries the ordering by itself**, and the 5-against-6 is illustration.

That is the operational content of the decomposition and what a practitioner should take from §5.

One clarification the title invites. "Selection, not capacity" contrasts with the *character budget*, which was never binding: the target cost 5,058 of 32,000 characters while deployed selection spent 31,946 to deliver a third of it. The candidate pool is a capacity limit of a different kind — on how many episodes the selector may consider — and it binds first. Both hold. What the program spent four studies believing, and what is false, is that the character budget was the constraint.

We have not found this decomposition reported elsewhere in this space. The reason is likely mechanical: computing it requires a per-fact known optimum on the same store, which requires both an answer key and exact-cost accounting, and evaluations reporting end-to-end scores against a benchmark do not usually carry either.

6. What survives

The 12 of 17 is offline availability, and it has now been tested live — it did not promote. LV-001 ran the shipping configuration against the deployed baseline over this corpus at one seed. The six-item offline advantage produced a one-item live difference in correctly attributed answers, and the configuration scored **2.0 lower on targeted probes** than the baseline it was meant to replace — failing its own pre-registered no-regression bar by four times the tolerance. Its registered status is **not promoted**.

What survives below is unaffected. The component guarantees are tested, and §5′s decomposition concerns which episodes reach the window, not what a model does with them once they are there. What LV-001 removes is the assumption that the second follows from the first (§5.1.1, §8.6).

6.1 What was removed

Each mechanism below was built, measured, and closed by the result beside it. They do not correspond one-to-one with the studies: several fell to the same study, and some outlived the study that first weakened them.

Removed	Killed by
Distillation and "dreaming"	Five studies; query-blind selection cannot anticipate a later query
Promotion filters	The weighted route was arithmetically unreachable; every promotion came via bypass

Removed	Killed by
Topic layer and consolidation	52 topics for one 120-turn conversation; 12 domains collapsed to 2 at 1,000 turns
Associative graph from co-activation	No configuration cleared its advancement gate
Query-type routing	Oracle ceiling 6.09% against a 10% threshold
Approximate nearest-neighbour search	Recall degraded at synthetic scale
Query segmentation	Improved its matched-budget baseline from 6/17 to 10/17 and still failed its locked 14/17 bar
Attention-derived term selection	Run as an oracle over 714 candidate cue rows: 0 reached the retrieval threshold
Entity extraction as primary index	Zero entities in the target span
Density for formation	Ranks the six hardest facts 89th–316th
Inverse document frequency for formation	Three unregistered variants disagree; no family-level negative result
Rule detection and persistence	Failed at 1,000-turn scale

6.2 What remains

An append-only verbatim store. A recency window. Cosine-threshold similarity retrieval for targeted queries. A set-level coverage objective for selection. Everything packed at exact serialized cost against one budget.

The recency window in that list is the component's own, and it is a genuine last-N window. The live studies that produced the results above ran two other rules under the same name: a least-recently-delivered rotation over the whole store in the most recent study, and before it a block that locked onto the conversation's first nine turns and held them. §5.2.4 and §5.2.5 give the measurements and the consequences.

One formerly open item is no longer owed as a component mechanism. The component emitted no absence signal on any of 500 EC-001 questions, while the fixed reader correctly abstained on 17 of 20 registered abstention items. This does not give the component an absence detector or establish reader robustness across models and prompts. It does show that component-level detection is not necessary for end-to-end abstention under this tested reader, so F3 is retired as a component requirement rather than marked solved.

There are no generative model calls anywhere in the memory path. Nothing in it asks a model to write text about the store. That is the property Study 005 established and the one every removal preserved.

It is not the absence of model calls. `context()` embeds the query on every call, and `append()` embeds every episode, so an embedding model must be resident. What follows from the architecture is that the memory path emits no generated text, and that its output is reproducible **given a pinned embedder** — which is why the library asserts a sentinel vector hash on every store open rather than assuming one (§7.4).

That is the whole architecture, and it is smaller than what this program started building. It contains none of the mechanisms listed above. We make no claim about how it compares to systems that were never run here.

6.3 Why a practitioner should care

Interpretation, not measurement. The properties are measured; the causal story about why they hold is a reading of this program's history.

The surviving design is reproducible and provenance-preserving, and both track a negative result rather than a design goal. It is reproducible because distillation was removed, and distillation was the component whose output could vary run to run: `context()` is a pure function of store state, query, and budget, verified byte-identical across two processes. It preserves provenance because every delivered character is a stored episode verbatim — there is no generated text about the store to be wrong, which follows from removing the mechanisms that generated such text.

Stated plainly, as a reading rather than a result: **the properties that make this component deployable were bought by the failures, not by the design.** A program that had succeeded at distillation would have shipped something harder to test and harder to audit.

The extraction is certified rather than assumed: all 132 committed selection records and all three committed rendered blocks reproduce their SHA-256 byte-for-byte through the library, with the full suite at 1,007 tests.

6.4 What it costs

These results come from the 1,000-turn endurance run, not the 121-turn corpus §5 uses. Nothing here establishes a §5 result at 1,000 turns, and nothing in §5 establishes these at 121.

Three things could grow as a conversation lengthens. Only one binds.

Delivered context is bounded, because it is enforced. Replaying the 1,000 committed episodes through the library at a 32,000-character budget, the delivered block breaches the budget on 0 of 1,000 turns and its 95th percentile moves +18 characters across the final five 100-turn buckets. The same block also **truncates on 895 of those turns**, dropping up to 70 episodes and wanting up to 65,864 characters. It is bounded because a ceiling binds during selection, not because demand is small. Both readings belong together; the first alone would be the kind of surrogate this program keeps catching.

Disk is cheap. 4,743 bytes per turn at the margin, 86% of it embeddings. About 48 MB at 10,000 turns. Nothing there ends continuous operation.

Latency binds. 190 ms at 1,000 candidates, clustering 81% of it and rising. The stated horizon is comfortable to a few thousand episodes and unusable in an interactive loop somewhere before 10,000. Figure 5, right panel.

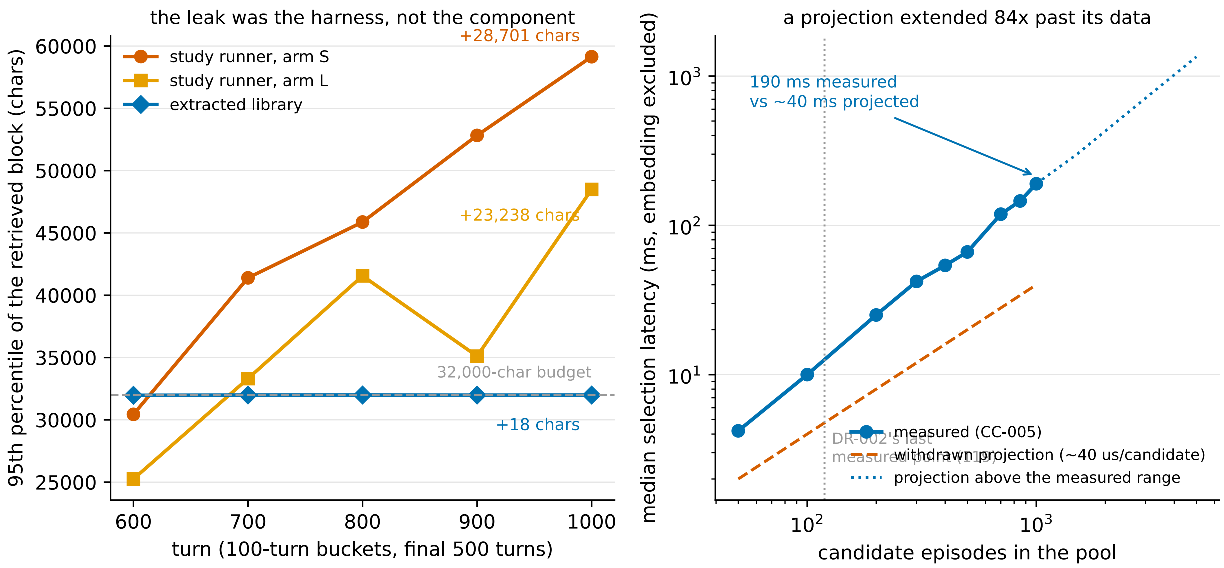

Figure 5. Growth and cost. *Growth belonged to the harness; cost was five times the projection.* Left: the 95th percentile of the retrieved block per 100-turn bucket over the final 500 turns of the 1,000-turn run. In the study runner the block rises 23,238 characters in one arm and 28,701 in the other, still setting records in the last bucket; replayed through the extracted library at the same budget it moves +18 characters and breaches the budget on 0 of 1,000 turns — while truncating on 895 of them, so it is bounded because enforced, not because demand is small. Right: measured median selection latency against candidate count with the withdrawn linear projection overlaid; 190 ms measured at 1,000 candidates against about 40 ms projected, exponent 1.25 over 50–1,000 where the earlier sweep found 0.96 over 20–119, and clustering's share rising from 37% to 81%. Values above 1,000 candidates are projections, drawn dashed. Sources: dx002_results.json f8ab79ab041cb3e3, ge0_growth_gate.json 350fe20bbc3beed9, latency_curve.csv 0d2b8075ff5a971f, growth_measurement.json 20d65a018e28b03f.

Those milliseconds are one machine's. The runtime is llama.cpp with a 27B generation model at UD-Q6_K_XL, one slot, fixed seed, speculative decoding disabled, and Qwen3-Embedding-0.6B over SQLite with sqlite-vec. Selection timings exclude embedding entirely — query and episode vectors are already resident — and are medians over repeated runs on a single machine. Only the exponent and the clustering share plausibly transfer.

The obvious fix — keep the pool small by dropping low-similarity episodes — is the one operation this program measured to break retrieval (§5.2). So retention is unbounded by policy, the trimming knob carries an unsafe_ prefix and the finding in its docstring, and the horizon is stated rather than engineered around.

7. Self-audit and corrections

Everything in §5 rests on this program's own measurements, scored by its own raters, against its own rubric. The reason to extend it credit is that the program audited itself and published what it found. All six items were caught by gates the program wrote, and all six are in ERRATA.md.

7.1 Four corrections, in brief

The scoring audit removed the program's only success. A blind re-scoring of 222 committed items across Studies 001–009 changed 19. Study 002′s iterative arm fell 13.0 → 8.5, because a truncated reasoning block had been credited as a complete response; Study 001 lost the program's only VALIDATED verdict. The residual estimate is extrapolated, and its precision is reported with it: 3 disagreements in a 26-item control sample projected across 143 unreviewed items gives 16.5 expected errors, but the 95% Clopper-Pearson interval on 3 of 26 runs from 2.4% to 30.2%, admitting **anywhere from about 3 to about 43**. A paper whose §7.3 is about an interval being misread should not present one number where the data supports a range that wide.

Every study on record ran over its stated budget. Study 010 reported two retrieval blocks at 31,991 and 31,847 characters and called them near-saturation of a 32,000 budget. Replayed character-for-character, their serialized lengths were 53,726 and 53,839 — **67.9% and 68.2% over**. The old accounting counted source text and omitted per-episode tags, metadata, and separators. Scores did not change, but they describe a budget-noncompliant arm, and the compact-store conclusion built on the undercharged figures was withdrawn.

A validator invalidated a published curve. A probe-order check verifies mechanically that every rubric-required fact is planted strictly before the probe asking for it. It found degradation probes requesting facts not yet planted, invalidating a published curve. The check now blocks artifact lock.

A projection ran 84× past its data. A pre-registered budget quoted about 40 microseconds per candidate at exponent 0.96 and projected 40 ms at 1,000 candidates. The source sweep covered **20 to 119**. Measured at 1,000, the cost is **190 ms, about five times the projection**, at exponent 1.25. The correction is to the range attributed and the extrapolation built on it, not to the original measurement. Figure 5, right panel.

7.2 The same query text returns different vectors

DX-001′s replay gate failed on its first attempt. The cause was not the query text but the shape of the embedding call: E005 embedded nine probe queries in one batch, the replay embedded one alone. The two vectors agree to a cosine of **0.999837**, with a largest single-component difference of **0.217**, and that difference flips **6 of 146** committed selection payloads.

A 0.999837 agreement reads as identical and is not. Reproducing a retrieval result requires reproducing the call shape, not only the text. The shipped library now embeds a fixed sentinel under the pinned call shape on every store open and asserts its hash against the one recorded at first open; drift raises rather than warns.

This has one direct consequence for reproduction. The 12-of-17 result was produced with the breadth query embedded in a nine-query batch; the shipped `context()` embeds a single query alone. The primary configuration is **not** among the six payloads that difference flips, so the headline reproduces under either — a checked fact rather than a safe assumption, and six other configurations do not have it.

It is also evidence about embedder sensitivity generally, which §8.4 collects.

EC-002 exposed a second vector-level limit: recomputing nominally identical solo-call embeddings moved one evidence rank and one coverage selection because EC-001′s original cache had not been retained. Its A0 is therefore an amended reproduction under recomputed embeddings, not a byte-exact replay. EC-001′s headline numbers reproduce in aggregate under that gate but remain permanently unreplayable at bit granularity. The library now persists exact float32 vectors, binds them by both file and canonical text-to-vector hashes, and makes a read-only miss fatal. That contract protects retained caches going forward; it does not repair the historical EC-001 record.

7.3 A diagnostic written to catch surrogate failures nearly committed one

This is the most instructive of the six.

The recurring failure class this program tracks is a check that can pass while the property it certifies is false. DX-002 was written to determine whether a 1,000-turn run's context was still growing. Its decision rule was a three-clause conjunction; its implementation checked one clause — whether the terminal slope's 95% confidence interval contained zero.

It did, for every component of the prompt, in both arms. The diagnostic returned "bounded".

The third clause was *no unbudgeted component climbing*, and one was. A block whose 95th percentile rose from 25,253 to **48,491 characters** across the final five buckets, still setting records in the last bucket, was reported as flat. The interval was wide because the series are sawtooths with autocorrelated residuals; it was measuring statistical power and was read as evidence of boundedness. The smallest slope the data could distinguish from zero was about 17 characters per turn — 17,203 characters of drift over 1,000 turns the fit would not have caught.

The rule was replaced with two readings assuming nothing about noise: whether the final bucket still holds the maximum, and how the terminal window compares against the one before it. The verdict flipped, and the near miss was written into the decision record rather than quietly repaired.

The finding that followed is Figure 5: the leak belonged to the study runner, which carried the recency window and the retrieval tier on separate budgets, not to the extracted component, which routes both through one.

7.4 What the gates missed

Every error above was caught by a gate this program wrote, which invites the obvious question and deserves the honest answer.

Between about 3 and about 43 scoring errors are estimated to remain unreviewed, point estimate 16.5. Study 010 was outside the scoring audit entirely, so its exploratory scores are not comparable to the corrected series. The mechanism seal for one tier was computed over mixed line-ending representations and referenced a database file never committed. And runtime independence was never measured: every number here comes from one model at one quantization on one machine, with §7.2 giving positive reason to expect a different embedder would move the §5 results.

7.5 A probe that removed the thing it was built to find

Amendment 001 raised a determinism gate failure: the same byte-identical prompt had produced two different answers at a fixed seed. Phase 1 was built to characterize that. It issued 20 committed windows ten times each under three sampler conditions, plus two conditions added beyond the registration, plus twenty replays of the exact prompt whose divergence was recorded. **820 generations, zero divergence.** The probe concluded the runtime reproduced and the recorded failure was an outlier of unidentified cause.

Phase 2 then ran the deployed configuration five times end to end and reproduced the divergence on the first turn of the second replicate: 343 characters against 80, diverging at character 79, matching both committed response digests exactly.

The probe missed it because it isolated the model call from the system that makes the call — no store, no embedding model, no 121-turn sequence, no alternating process lifecycle. It measured the component and reported on the system. That is this program's recurring surrogate failure class, arriving in a diagnostic written by an agent who had spent the previous day documenting the same failure class in three other places (§5.2.4, §5.2.5, §7.3).

The Phase 1 report is superseded in part rather than rewritten, its numbers left standing and its conclusion marked wrong, and its own surrogate-audit table now carries the row that failed.

8. Limitations

Each item names what would settle it. §5 states its own scope where it matters; this section is the complete list rather than a restatement.

8.1 The instrument's band is 3.0, and most scored comparisons here are below it. Every comparison in this paper is a single run at a fixed seed. That was recorded as a missing variance estimate until the estimate was made.

Five replicates of the deployed configuration — identical corpus, settings, seed and standing runtime, run back to back in one server process — scored **8.0, 8.0, 8.0, 8.0 and 11.0**. Max minus min is **3.0** against a decision rule committed before the replicates ran. It is not a spread but a switch: four of the five are byte-identical across all 121 turns, and the fifth, the only one to meet an empty server slot, diverges at turn 1 and never re-converges. Rater disagreement was measured separately and is near zero (64 of 65 items unanimous), so this is run-to-run variation and not scoring noise.

Applied uniformly and in both directions, the memory-tier contrast (3.0), the live-validation targeted regression (−2.0) and the tier-isolation kill (−1.0) are all **inside the band and not demonstrated**; the corrected treatment series (3.5) is the only scored result that exceeds it, and exceeding a band is not the same as being demonstrated. *Not demonstrated is not refuted* — these may be real, and a single run per arm cannot say.

What the band does not touch is everything offline and deterministic: gate outcomes, delivery counts, character accounting, packing measurements, EC-002′s 152 gains and zero losses, IC-001′s zero K deliveries at 8 of 8 probes, and the N-tier replays. Those are counts and identities, not scores, and §5′s decomposition rests on them. *Settled by:* repeated runs at multiple seeds with process state pinned, which no study in this arc pinned.

There is still no meaningful significance test. EC-001 ran cleaned LongMemEval-S, removing the claim that every result is self-authored, but API unavailability replaced the pinned benchmark evaluator with Phi, Mistral, and hosted GPT-5.4 raters plus hosted GPT-5.5 AI adjudication. Its 20.0% equal-quota and 12.22% post-stratified results are Codex-substituted integrity scores, not official or benchmark-comparable LongMemEval scores. LoCoMo remains unrun. Boundedness claims are statements about a 1,000-turn horizon and say nothing about 10,000. *Settled by:* repeated runs at multiple seeds and the official evaluator on the registered answers.

8.2 Literal enumeration still rests on a single probe. Every internal breadth number here — 6 of 17, 12 of 17, 14 of 17, and the rank-87 reading — comes from one question. EC-001 adds 121 multi-session reasoning questions as the registered closest analogue: any-turn availability is 24.0%, and the equal-quota Tier 2 subset scores 0 of 20. That establishes external synthesis weakness under this configuration, but not that all such questions are enumeration or that the internal four-domain rubric generalizes. *Settled by:* multiple literal enumeration probes across external domains.

8.3 AI raters, AI adjudicators. Scoring used three blind passes with registered adjudication triggers, but the adjudicators were subagents, not humans. The control sample disagreed at 11.54%. *Settled by:* human adjudication of the same items.

8.4 One runtime, and positive evidence of fragility. One model, one quantization, one machine, one embedder. Whether §5′s results survive a *different* embedder is unmeasured — but this is not a blank absence of evidence, and it would be convenient to call it one. §7.2 shows the *same* embedder, given the same text under a different call shape, returning a vector agreeing to cosine 0.999837 that flips 6 of 146 committed payloads. A perturbation far smaller than a model change already moves 4% of results, so the reasonable prior is that §5′s specific numbers are embedder-dependent. *Settled by:* rerunning the E005 sweep under a second embedder.

8.5 Planted facts do not represent the dominant external ranking pattern, and packing causally suppresses external delivery. The corpus is constructed, and §5.5.1 gives a specific reason to suspect the inversion is a property of the planted vocabulary. RD-001 could not identify that mechanism: unchanged rarity scores cover only 6 of 76 fact-bearing episodes, across three variants with no registered primary or episode aggregation. EC-001 tests the outcome on a corpus this program did not construct. Only 69 of 470 answerable questions (14.7%) put all evidence below the top four, with median evidence-session rank 2. The dominant-generalization claim is therefore rejected. Of the 401 questions with evidence in the top four, 305 still retrieve no evidence session under EC-001. EC-002 holds the store, vectors, threshold, selector, and budget fixed and reverses packing priority. Any-session recall rises 109/470 -> 261/470, with 152 gains and no losses; exact-turn-any rises 79/470 -> 196/470, with 119 gains and two losses. All 500 blocks remain truncated, their median composition is 16 recency, 0 non-recency K, and 1 coverage exchange, and 91 of the 109 session hits come from recency. The external calibration therefore strengthens the corpus caveat while the counterfactual isolates one downstream cause: naturalistic questions do not show the same dominant inversion, and N-first exact packing suppresses delivery. The cause of the ranking difference remains open, as do the residual effects of threshold and exchange/session granularity. *Settled by:* an explicitly post-rank RD-002 rarity design; separately registered threshold and granularity counterfactuals; and a prospective live reader test of K-first packing.

8.6 Availability is not correctness, and the known optimum is mostly prior answers. §5.1.1 in full: four of five optimum episodes are prior probe exchanges, this probe's earlier answers were largely wrong, and an item counts as available if its text appears — however wrong the surrounding response.

This was the paper's largest structural weakness. **It has now been measured, and the weakness is real.** LV-001 pre-registered a two-arm live run of the shipping configuration against the deployed baseline over this corpus at one seed, fixing its thresholds and both reporting outcomes before any number existed. It ran, and it returned:

Registered bar	Result
B1 — does the offline advantage convert?	**WEAK.** Six-item offline gap became +1 correctly attributed item, inside the band the design called noise
B2 — no targeted regression, tolerance 0.5	**FAIL.** The shipping configuration scored 1.5 of 8 against the baseline's 3.5, a 2.0 shortfall

B2 was registered as a kill: *"A B2 failure kills the promotion regardless of B1."* The configuration's status is therefore **not promoted**.

The mechanism is legible. Asked for the two formatting rules planted in turns 1 and 2, the shipping configuration reported that it could not see the start of the conversation — the coverage objective had spent its budget on domain spread and stopped carrying the opening, which per-item cosine ranking had retained. Offline, that same configuration preserved 16 of 16 targeted items. **Preserving an item's availability and preserving the answer that depends on it are not the same property**, and this program had measured only the first.

One run, one seed, one rater where the protocol asks for three. The −2.0 is as unreplicated as the +1. Full detail, including the fabrication both arms produced on the domain neither retrieved, is in experiments/components/live_validation/LV_001_report.md.

8.7 Amendments exist after results. Twelve in the bakeoff alone. The program records per amendment whether it preceded the result it affects, and applies a legitimacy test permitting corrections to measurement units and protocol contradictions while forbidding making a criterion easier once results are known. The record is published so a reader can disagree with individual calls.

8.8 Figure 3 now has the full ordering; the rarity join still fails. When the paper was drafted, per-episode cosine ranks were committed for 16 of 119 candidates. RD-001 recovered all 119 under E005′s pinned nine-query call and replayed those 16 checks with only the already-published turn-118 correction from rank 21 to rank 20. Figure 3 now plots the full ordering. That recovery does not repair the proposed rarity test: the prior audit scores six source episodes, not the 76-episode population, so no correlation is reported. A provenance audit also finds that no IDF variant was primary: only mean IDF ranks all five eligible hard-plant spans worse than density, while maximum and summed-per-word IDF each improve at least one.

8.9 External scoring is substituted, not official. EC-001′s equal-quota subset scores 28 of 140 (20.0%), and post-stratification to the 500-question population gives 12.22%. Amendment 010 forbids placing either number directly against published LongMemEval results because the pinned GPT-4o evaluator was unavailable. Three blind family passes and 20 independent AI adjudications improve integrity over a single pass, but GPT-5.4 and GPT-5.5 are hosted display selections without immutable API snapshots, and the adjudicator is not human. The full boundary is in experiments/external/longmemeval/EC_001_REPORT.md.

9. Conclusion

Eleven pre-registered efforts on one program produced one architecture worth keeping and a measurable account of why the rest did not work.

For a practitioner, the useful part is the subtraction. Every mechanism the program built failed its own gate; what is left is an append-only verbatim store, a recency window, similarity retrieval, and a set-level coverage objective, with no generative model calls in the memory path. That component is reproducible given a pinned embedder and auditable line by line, and §6.3 argues both properties followed from the removals rather than from foresight. Read §5.2.4 and §5.2.5 before carrying that list across: the component's recency window is a genuine one, and no live study behind these results ran it — they ran a rotation over the whole store, and before that a block frozen on the conversation's opening. Whether a real window would do better is not something this program measured. If a memory component in your system makes generative calls, this program's experience is that they bought less than they cost — on one internal corpus, with one later external stress test that is not a published-system comparison.

For a researcher, the useful part is the decomposition and the order inside it. Retrieval failure here was not one thing. The candidate pool decided what could be seen, the objective decided what was worth taking, and a similarity floor decided what was unreachable at any weighting. The order is forced rather than tidy, and structurally so: one of the four domains has no representative anywhere in the deployed shortlist, so no objective can recover it from there. Widening the pool is not the first fix because it measured better. It is first because the alternative is impossible. Separating the three required a per-fact known optimum on the same store — a measurement costing an answer key and exact cost accounting, and buying a sharper question than an end-to-end score.

The observation was tested elsewhere and narrowed. On the internal corpus, for the one enumeration probe this program has, the four highest-cosine episodes carried none of the target facts and the last needed item sat at rank 87 of 119 — while the same ordering placed every needed item inside rank 2 on all eight lookup probes. On cleaned LongMemEval-S, only 14.7% of answerable questions put all evidence below the top four, and multi-session questions fail end-to-end without showing a stronger inversion. The internal result remains real, but it is not the dominant external pattern. §5.5.1 records why neither the failed rarity join nor this multi-factor external comparison identifies whether planted vocabulary caused the difference.

The program's own summary of eleven efforts is that the model used what it received. At the hardest probe it used all ten available facts and invented none. The failures were delivery failures, and delivery turned out to be a selection problem sitting on top of a candidate set already narrowed by the wrong rule.

Appendices

- **A. Claim-to-artifact table** — paper/CLAIM_TO_ARTIFACT.md. Every claim with its committed artifact and hash; two claims cut or demoted for want of one.
- **B. Study table** — §4 above; full reports under experiments/study_NNN/.
- **C. Amendment record** — per-study amendments/ directories, each with its before/after-result status.
- **D. Corrections index** — ERRATA.md, cross-referenced from §7.
- **E. Reproduction** — paper/REPRODUCTION.md and paper/reproduce_headline.py. Rebuilds the 5,058-character exact optimum from the committed turn log through the installed library and checks its length and SHA-256 against AR-001′s committed values. Verified in a clean virtual environment containing only episodic. The selection results cannot be reproduced this way, because the store's vectors were never committed; the appendix says so.
- **F. Spec reconciliation** — paper/notes/EVIDENCE_INDEX.md §1, recording six places where the authoring specification and the committed artifacts disagreed and the artifact won.
- **G. Review record** — paper/reviews/: two adversarial cycles, a slop audit, and the three-reader readability review that prompted this restructure.

Typeset PDF. paper/Selection_Not_Capacity.pdf, built from this file by scripts/build_paper_pdf.py (pip install typst). The PDF has no independent source: it is generated from this Markdown, figures are placed at the paragraph that first cites them, and figure numbering comes from here rather than from the typesetter, so the numbers cannot drift from the prose. If the two ever disagree, this file is right and the build script is broken.